

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



Deep Learning

0. Logistics

Spring 2026

Adapted from CMU 11-785 slides:

<https://deeplearning.cs.cmu.edu/F25/index.html>



Introduction



Neural Networks are taking over!

- AI method **inspired by the human brain** that teaches computers to **process data in a way that is inspired by the human brain**
- Widely used in **pattern recognition**, prediction, and **analysis problems**
- Established **state-of-the-art** in many problems, often exceeding the previous benchmarks by large margins

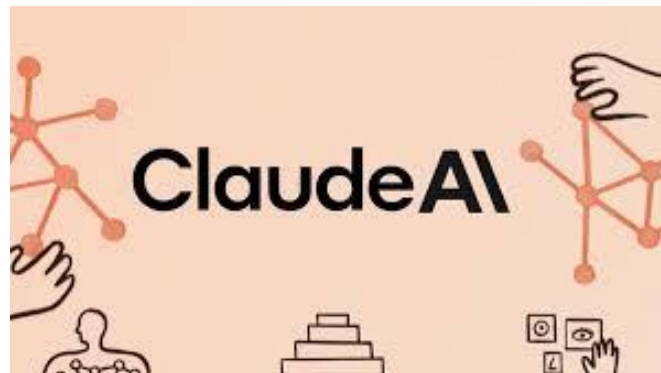
Breakthroughs



OpenAI's conversational model



predicts 3D structure of proteins

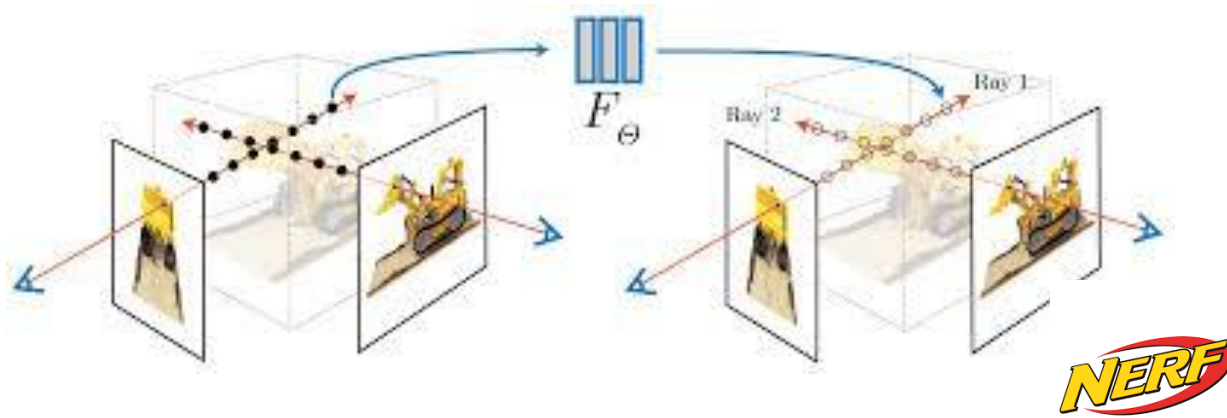


Anthropic's chatbot



creates images from text descriptions

Breakthroughs



generates photorealistic 3D scenes from 2D images



robotaxi service- self-driving cars



The Instructor, Course Topics, and You



Instructor

- Instructor:
 - Kashif Javed

- Office hours:
 - Tuesday, 11 am to 12:30 pm
 - Location: Chairman's office, Department of Computer Engineering



Who is the course designed for?

- Required:
 - Python
- Highly Recommended:
 - Calculus (differentiation, chain rule), linear algebra, and basic probability
- Nice to have:
 - Vector Calculus
- Forgotten anything?
 - https://d2l.ai/chapter_appendix-mathematics-for-deep-learning



General Course Topics

- Introduction and preliminaries
- Neural Networks for Regression/Classification
- Multilayer Perceptrons
- CNNs/Modern CNNs
- RNNs/Modern RNNs
- Attention Mechanisms and Transformers
- LLMs/GANs

Basics/Preliminaries

DL Techniques

Applications



Who is the course designed for?

- This course is implementation heavy
 - There is a lot of coding
 - There is a lot of experimenting
 - You will work with some large datasets
- Course **Language** of choice: **Python**
- Course **Toolkit** of choice: **PyTorch**
 - You are welcome to use other languages/toolkits



Schedule

- Courses AI-503 and EE-550:
 - Lecture: Tuesday and Thursday , 4:30 p.m. – 6:00 p.m.
 - Location: EE-125A, Department of Electrical Engineering

- Courses CMPE- 441 and CMPE-642:
 - Lecture: Monday, 9:00 a.m. – 12:00 p.m.
 - Location: CE 110, Department of Computer Engineering



The Challenge of the Course



This course is not easy

- A lot of work!
 - A lot of work!!
 - A lot of work!!!
 - **A LOT OF WORK!!!!**
-
- But a lot of fun :)





Mastery-based evaluation

- Quizzes/Mid/Final Exams to **test your understanding** of topics covered in the lectures
- HWs to teach you to **implement complex networks** and optimize them to high degree
- Project to **expose you to real world DL problems**



Components of the Course

Lectures



- In-class lectures



Quizzes

- Questions are related to topics covered in lecture slides



Assignments

- There will be FOUR main HWs
- Main HWs all have two parts:
 - Part 1: Autograded problems with deterministic solutions
 - Part 2: Open problems posted on Kaggle



Course Project

- Projects are intended to exercise your ability to comprehend and implement ideas beyond those covered by the HWs

- Projects can range from
 - *Verification* of results from “hot” published work
 - *Implementation and evaluation* of cutting-edge ideas from recent papers
 - Problem-solving that might lead to publication, if completed well



“Go the extra mile”



Lecture Attendance

- Our performance metrics over the semesters show a distinct correlation between attendance and course scores
- We also note a distinct inverse correlation between attendance and the amount of help you require

Recitation 0



- Go through the videos for Recitation-0 (0.1, 0.2, 0.3) and complete the notebook exercises
 - Gain comfort with the coding required in the HWs, early on



Reading Materials

- Zhang et. al. “Dive into Deep Learning”, Camb. Uni. Press, 2023
 - <https://d2l.ai/index.html>
- Bishop and H. Bishop, “Deep Learning: Foundations and Concepts”, Springer, 2024
- S. Prince, “Understanding Deep Learning”, The MIT Press, 2023



Reading Materials

- Géron, “Hands-On Machine Learning with Scikit-Learn and PyTorch”, O’Reilly Media, Inc. 2026
- I. Pointer, “Programming PyTorch for Deep Learning”, O’Reilly Media, Inc. 2019
- 11-785 Introduction to Deep Learning
 - <https://deeplearning.cs.cmu.edu/F24/index.html>



Components of the Grade



Evaluation

- Scores are evaluated based on 3 types of tests:
 - Written exams
 - Quizzes (not for AI-503, EE-550 and CMPE-642 students)
 - Mid-term and Final exams
 - HW Assignments (only first 2 assignments for CMPE- 441 students)
 - Project



Course Project

- **Teams:** think about forming project teams as soon as possible
 - CMPE- 441 teams are of 2-3 students maximum
 - AI-503, EE-550 and CMPE-642 teams are of 1 student maximum

- **Project requirements:**
 - Submit a project **proposal** the **end of week 6**
 - Submit a project **mid-term report** by **end of week 11**
 - Submit a final report 3 days **before the viva-voce due date**
 - viva voce will be held during the **last week of the semester**

Grading



■ <u>Written Examinations</u>		75
1 Mid-term Exam		30
1 Final Exam		40
2 Quizzes		5
■ <u>Assignments</u>		10
HW1		5
HW2		5

Grading



■ Project	15
Proposal	-
Mid-term report	5
Full report	5
Viva-voce	5



Homework Deadlines

- There are separate deadlines for Part-1s and Part-2s



Integrity



Personal and Academic Integrity

- You are **here to learn DL yourself**, not to demonstrate how well you can pretend to learn
- You are **at UET** – which means you are among the brightest and best students in Pakistan
 - You probably were among the top students in your peer group all your life, before you came here
 - It will be an **insult to yourself, to lower your integrity**
- **Try your best, honestly**. If you are unsure about something, reach out to me



Integrity

- Every student must solve every homework by themselves
 - You may discuss the homeworks with your friends, and even help them debug their code, but when you finally solve it, every line of your code must be written by you
 - Your solution must be yours
- Plagiarizing code from the web or your friends constitutes an academic integrity violation
 - If found guilty, minimum penalty is 0 marks for the work being graded
 - Maximum penalty would be limited by UET's plagiarism and cheating policy



Thank you for deciding to take
this journey of Deep Learning
with me

Prof. Kashif Javed